OpenADMET CYP Challenge — Time-Dependent Inhibition Track
Model Description Report

Team / leaderboard alias: Cheminfo
Predicted properties: `CYP3A4_is_TDI`, `CYP2D6_is_TDI`
Training data: challenge-distributed data only; no external TDI corpus was merged

1. Reformulating the problem: we do not classify the label

The central design decision, from which every architectural choice below follows, is that **we never train a classifier on `is_TDI`**. The published label is not an annotation to be learned — it is a deterministic function of two measured potency surfaces:

```
is_TDI  ==  max(pIC50_TDI_condition, floor) - max(pIC50_direct_inhibition, floor)  >
log10(2)
```

We verified that this rule reproduces the published label exactly on every training row carrying both surfaces. Treating the problem as binary classification therefore throws away two things the data actually contains.

First, it discards distance from the decision boundary. The floored shift is a continuous, physically meaningful quantity — the degree to which pre-incubation potentiates inhibition — defined wherever both surfaces exist. A compound that shifts just below the two-fold cut and one that does not shift at all are indistinguishable to a classifier and entirely different to a regressor. Our primary heads therefore **regress the floored shift** and the binary call is recovered by applying the challenge's own threshold to the regressed value. The gain from this reformulation, rather than from any choice of learner, is the single largest improvement we obtained on CYP2D6.

Second, it hides that the CYP3A4 label is a conjunction. A substantial block of CYP3A4 rows have a fitted TDI-condition curve but no fitted direct-inhibition curve, and the organisers' convention assigns every one of them a negative label regardless of chemistry. Whether a direct curve could be fitted at all is thus a constituent of the label, not a missing-data nuisance. We model it explicitly, as described in Section 4.

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2. Architecture A — multitask directed message-passing network

The first family is a directed message-passing neural network (D-MPNN) operating on the molecular graph, with a shared trunk and multiple output heads.

Representation. Messages are passed along directed bonds for several rounds and aggregated to a learned graph-level embedding. That embedding is concatenated with a block of computed two-dimensional physicochemical descriptors before the feed-forward head. The concatenation matters for this endpoint: bioactivation propensity depends on global molecular properties — lipophilicity, polar surface, charge distribution — that a bond-level message passing scheme reconstructs only indirectly, while a hashed substructure representation dilutes them. Descriptors that are degenerate or numerically unstable across the training set are removed, and the retained block is standardised and clipped so that a handful of extreme values cannot dominate the shared trunk.

Multitask structure. Both scored isoforms are learned jointly on one trunk, with additional auxiliary heads predicting whether each isoform's potency crosses the activity threshold implied by the label definition. The auxiliary heads carry reduced loss weight. Their purpose is regularisation of the shared representation rather than prediction: they force the trunk to encode absolute potency, which the shift alone does not require it to do, and which the label definition makes relevant through the flooring operation.

Ensembling. The final model is an ensemble of independently initialised replicates trained across several random seeds, averaged at the prediction stage. Cross-validation throughout uses Bemis–Murcko scaffold folds rather than random folds, so that model selection is never informed by scaffold memorisation — an important control here, because the training set contains congeneric series and the blind set is a screening collection.

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3. Architecture B — gradient-boosted trees over stacked endpoint surfaces

The second family deliberately abandons a learned representation in favour of an explicit, high-dimensional feature table consumed by gradient-boosted decision trees. It is not included as a baseline; it wins one of the two isoforms outright and is a genuinely different inductive bias.

Feature construction. Three blocks are concatenated:

1. **Physicochemical descriptors** — the same computed property block used above.
2. **Substructure counts** — circular fingerprints in both a standard connectivity flavour and a pharmacophore-typed (feature-based) flavour, taken as *counts* rather than bits, plus a public structural key set. Counts are used because a boosted tree can split on multiplicity, which a binary bit vector destroys.
3. **Stacked endpoint surfaces** — described next, and the block that carries the model.

Stacked out-of-fold surfaces, including unscored isoforms. This is the most distinctive element of the second architecture. Rather than feeding only structure, we first fit separate regression models to each underlying potency surface — the TDI-condition potency, the direct potency, and the derived shift — for *all four* CYP isoforms present in the data, including CYP1A2 and CYP2C9, which the TDI track does not score. Each model's out-of-fold predictions become input features to the final per-isoform model.

The rationale is mechanistic. Time-dependent inhibition arises from metabolic bioactivation to a reactive species, and a compound's susceptibility to oxidative bioactivation is largely a property of the molecule rather than of the particular isoform that performs the oxidation. The unscored isoforms therefore carry real information about the scored ones, on compounds for which no scored-isoform measurement exists. This proved correct and non-trivially so: in the final CYP2D6 model, the most informative feature after CYP2D6's own shift surface is the predicted **CYP1A2** shift — an endpoint that is never evaluated. Cross-isoform bioactivation transfer is a real and exploitable signal in this dataset.

> **A negative result we consider worth reporting.** We added an explicit panel of SMARTS patterns for the classical mechanism-based-inactivation motifs — methylenedioxyphenyl, furan, thiophene, terminal acetylene, aniline, thiourea, hydrazine, cyclopropylamine, quinone and related groups — on the expectation that a tree model would split on them directly where a hashed fingerprint dilutes them across bits. They contributed essentially nothing: most carried zero split gain, the best ranked far below the structural and stacked blocks, and the panel as a whole accounted for around one percent of total model gain. Either the count-based fingerprint blocks already capture these motifs adequately, or time-dependent inhibition in this dataset is not driven by the textbook alert list. We record it so the experiment need not be repeated.

4. The conjunctive head for CYP3A4

Following from Section 1, CYP3A4 is modelled in both primary architectures as the conjunction the label actually is, rather than as a single surface. One head predicts the floored shift; a second, separate head predicts whether a direct-inhibition curve was fitted at all. The second head's output **gates** the first: a compound predicted not to yield a fittable direct curve is driven toward a negative call irrespective of its predicted shift, because the convention makes it negative by construction.

A corollary we tested and acted on: the label identity also implies that a positive call *requires* the TDI-condition potency to exceed the activity floor plus the two-fold shift. A regression on TDI-condition potency can therefore be used as an independent veto, demoting predicted positives that cannot satisfy the arithmetic. We emphasise that this is a consequence of the label's definition rather than a learned heuristic, and the direction is one-way — it can only remove calls that the definition forbids, never add any.

We also examined whether to model the conjunction *compositionally*, estimating the label as the product of the two heads' probabilities rather than estimating it directly with a gate. We did not adopt it. Being structurally correct about the label does not make a product of two noisy estimators better than one head trained on the conjunction directly: the two heads' errors multiply, and the direct estimator measurably outperformed the composed one. The decomposition is valuable as a diagnostic and as a gate; it is not a better estimator.

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5. How the blend was created

The two architectures are combined **per endpoint**, in **rank space**, with weights selected out of fold. Three deliberate choices define the blend.

(a) Rank space, not probability space. The two families produce scores that are not on a common scale and not mutually calibrated — one is a regressed potency shift, the other a boosted-tree score over a sparse feature table, and the subsequent gating operations are not probability-preserving. Averaging them as probabilities would let the more confidently scaled model dominate for reasons unrelated to its accuracy. We therefore convert each model's output to within-isoform ranks and blend the ranks linearly. This is also the correct space for the objective: at a fixed number of positive calls, the evaluation metric depends only on the ordering of compounds, so the blend operates directly on the quantity that is scored.

(b) Per-endpoint weights, because no family wins both isoforms. The two architectures have genuinely different strengths here, and a single global weight would be wrong for one of them. On CYP3A4 the graph network is the stronger ranker and is kept as the majority component, with the tree model contributing a substantial minority weight. On CYP2D6 the ordering reverses: the shift-regression formulation on explicit features is the stronger component and carries most of the weight, with the graph network contributing the remainder. Weights were chosen on out-of-fold predictions over the scaffold folds and then frozen; we did not tune them against the public leaderboard.

(c) A third family used as a filter, not as a predictor. Beyond the two-model blend we trained an independent third family following a published protocol for CYP inhibition prediction: a grid of classical learners (random forest, support-vector machine, nearest neighbours and a shallow neural network) over a range of public fingerprint types, with per-fold model selection and class-rebalancing variants, retrained on the challenge's own data. Model selection within the grid chose a random forest over a single fingerprint block for both scored isoforms.

This family deliberately does not use any of the reformulation described above. It classifies the published binary label directly — no floored-shift regression, no curve-fittability

head, no conjunctive gate, no stacked endpoint surfaces, and no physicochemical descriptor block. It is a plain binary classifier on a single substructure representation, and it is the only model we built that treats `is_TDI` as an annotation rather than as an arithmetic identity. Trained this way it is, unsurprisingly, a substantially weaker ranker than the blend, and we never submitted it as a standalone prediction.

That weakness is precisely why it is useful. Because it shares neither the representation, nor the learner, nor the target formulation of either primary architecture, its errors are substantially decorrelated from theirs — it is wrong about different compounds. Positive calls are therefore retained only where the blend and this independent family agree in the upper region of their respective rankings. The effect is a purity filter rather than a ranking change: it removes calls on which two methodologically unrelated models disagree, and a weak-but-independent opinion is worth more in that role than a strong-but-correlated one. We note that the retention threshold is itself a tuned quantity and that we found it to sit at an interior optimum — both tightening and loosening it degraded the result, on both isoforms.

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6. Choosing the operating point

Converting a ranking into binary calls requires a threshold, and the obvious approach — tune it on out-of-fold data — is wrong here, because the training set and the blind set have different class balances. The training collections are hit-enriched relative to a screening population, and substantially so for CYP2D6.

The blind set's prevalence, however, is recoverable analytically. Every leaderboard entry reports accuracy, precision and recall over the same compound set, and those three quantities over-determine the positive rate:

```
prevalence = (1 - accuracy) / (1 - 2*recall + recall/precision)
```

Solved independently across the public entries, this converges tightly and consistently for both isoforms. We therefore reweight our out-of-fold predictions to the blind set's prevalence before selecting the call rate that maximises the evaluation metric, rather than carrying over a threshold tuned at the training prevalence.

A bi-normal analysis of the metric clarifies what the resulting call rates mean. The optimal call rate is a function of ranking quality, not a universal constant: a strong ranker should call close to the prevalence, while a weak ranker should call well above it, and the optimum is flat over a wide range in between. This is worth stating because low call rates are sometimes assumed to be inherently correct; they are correct only in proportion to how well the ranking separates the classes.

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7. Structure preparation

All training and test structures were prepared with Schrödinger LigPrep, generating low-energy conformers with protonation states assigned at physiological pH. LigPrep output was then audited for systematic protonation and bond-order artefacts, which were corrected by custom cheminformatics scripts applied in a fixed sequence: neutralisation of anionic sulfonamide nitrogen, of pyridinium-type aromatic nitrogen, of protonated morpholine nitrogen, and of protonated amidine/guanidine/iminium nitrogen; and repair of deprotonated phenols and of inverted amide bond orders. Permanently charged quaternary centres are left unchanged, as are genuinely basic sites elsewhere in a molecule carrying a corrected sp^2 nitrogen. Identical corrections are applied to training and test structures.

8. Summary of what is novel here

1. **Label-identity-driven reformulation.** Recovering the exact arithmetic identity behind `is_TDI` and regressing the underlying continuous shift, rather than classifying the published binary.
2. **Explicit conjunctive modelling of CYP3A4**, with a curve-fittability head gating the shift head, and an arithmetic veto derived from the label definition rather than learned.
3. **Cross-isoform bioactivation transfer** through stacked out-of-fold potency surfaces from isoforms that the track does not score — which turn out to be among the most informative features available.
4. **Rank-space, per-endpoint blending** of two architectures with different inductive biases, operating directly in the space the metric scores.
5. **A deliberately weak, methodologically unrelated third model used as a consensus purity filter** rather than as a predictor — a plain binary classifier that keeps the un-reformulated label precisely so that its errors stay decorrelated from the primary architectures'.
6. **An operating point derived analytically from the public leaderboard's own published statistics**, correcting for the enrichment difference between the training and blind populations.

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9. Software

Graph neural network: ChemProp. Gradient-boosted trees: LightGBM, with XGBoost used for the label-arithmetic veto regression. Classical consensus family: scikit-learn over public fingerprint descriptors. Cheminformatics throughout: RDKit. Structure preparation: Schrödinger LigPrep. Training was performed on a single GPU instance for the neural ensembles and on CPU for the tree models.